Diego Menendez

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EDUCATION

University of California, Irvine 06/2027
Bachelor of Science in Aerospace Engineering
GPA: 3.98/4.00

RESEARCH EXPERIENCE

X Energy Lab — Advisor: Prof. Xian Shi 03/2024 - Present
Undergraduate Researcher

- Designed & fabricated a CH_4/O_2 rotating detonation rocket engine (RDRE) end to end — injector and manifold geometry, high pressure 20–400 g/s feed system, ignition system, data acquisition, engine control, and procedures
- Achieved sustained detonation for hundreds of seconds of cumulative runtime, characterized with high-speed imaging, chamber pressure, and laser Doppler vibrometry
- Developed cellular detonation structure analysis framework; results presented at IWDP 2026, Yokohama, Japan
- Ran narrow-channel detonation experiments with high-speed schlieren & shadowgraph imaging
- Funding affiliation NSF Grant #DMREF-2522673

UCI Rocket Project Liquids 06/2024 - 12/2025
Propulsion Engineer

- Redesigned the triplet impinging injector hole placement to address a center-faceplate burnthrough
- Selected injector materials and conducted water flow testing, and supported methalox hot fires as test-stand crew

UCI Computational Solid Mechanics Laboratory — Advisor: Prof. Julian Rimoli 01/2024 - 06/2024
Undergraduate Researcher

- Designed and built a linearly actuating testing machine to compress tensegrity structures
- Improved tensegrity structure fabrication method, reducing fabrication time by 50%

CONFERENCE PRESENTATIONS & PAPERS

- **Menendez, D. K.**, Shi, X. "Collision Family Framework for Cellular Detonation Structure." International Workshop on Detonation for Propulsion (IWDP 2026), Yokohama, Japan. May 2026. [Poster]
- **Menendez, D. K.**, Ross, M. W., Mumm, D. R., Valdevit, L., Shi, X. “Component and material failure mechanisms in miniaturized rotating detonation engines.” International Workshop on Detonation for Propulsion (IWDP 2026), Yokohama, Japan. May 2026. [Oral]
- **Menendez, D. K.**, Shi, X. “Numerical Simulation of Detonation Turbine Generator” UCI Undergraduate Research Opportunity Program Symposium (UROP 2025), UC Irvine. 2025. [Poster]
- **Menendez, D. K.**, Ross, M. W., Valdevit, L., Mumm, D. R., Shi, X., Dudko, N., and Vermaak, N. "Mapping Material Responses and Failure Mechanisms for a Heat-Sink Rotating Detonation Engine." AIAA SciTech Forum 2027. [abstract accepted].
- Ross, M. W., **Menendez, D. K.**, Shi, X., Mumm, D. R., Valdevit, L., Dudko, N., Vermaak, N., Yang, S. E., and Taheri-Mousavi, S. M., "Microstructural Influence on Material Degradation in a Rotating Detonation Engine," AIAA SciTech Forum 2027 [abstract accepted].
- Dudko, N., **Menendez, D. K.**, Ross, M. W., Yang, S. E., Taheri-Mousavi, S. M., Valdevit, L., Mumm, D. R., Begley, M. R., Shi, X., and Vermaak, N., "Experimental Investigation into the Thermomechanical Vibratory Response of Uncooled Rotating Detonation Engine Combustor Liners," AIAA SciTech Forum 2027 [abstract accepted].